AI ASSISTED CODING

LAB-13.1

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Batch-11

Task Description #1 (Refactoring – Removing Code Duplication)

• Task: Use AI to refactor a given Python script that contains multiple repeated code blocks.

• Instructions:

o Prompt AI to identify duplicate logic and replace it with functions or classes.

o Ensure the refactored code maintains the same output.

o Add docstrings to all functions.

• Sample Legacy Code:

```
# Legacy script with repeated logic
print("Area of Rectangle:", 5 * 10)
print("Perimeter of Rectangle:", 2 * (5 + 10))
print("Area of Rectangle:", 7 * 12)
print("Perimeter of Rectangle:", 2 * (7 + 12))
print("Area of Rectangle:", 10 * 15)
print("Perimeter of Rectangle:", 2 * (10 + 15))
```

• Expected Output:

o Refactored code with a reusable function and no duplication.

PROMPT:

#Identify duplicate logic and replace it with functions or classes, make sure the refactored code maintains the same output.

GIVEN CODE AND OUTPUT:

```python
#identify duplicate logic and replace it with functions or classes, make sure the refactored code mainta
def calculate_rectangle_area(length, width):
    """Calculate the area of a rectangle given its length and width.
    input: length and width of the rectangle
    output: area of the rectangle"""
    return length * width

def calculate_rectangle_perimeter(length, width):
    """Calculate the perimeter of a rectangle given its length and width.
    input: length and width of the rectangle
    output: perimeter of the rectangle"""
    return 2 * (length + width)

print("Area of Rectangle:", calculate_rectangle_area(5, 10))
print("Perimeter of Rectangle:", calculate_rectangle_perimeter(5, 10))
print("Area of Rectangle:", calculate_rectangle_area(7, 12))
print("Perimeter of Rectangle:", calculate_rectangle_perimeter(7, 12))
print("Area of Rectangle:", calculate_rectangle_area(10, 15))
print("Perimeter of Rectangle:", calculate_rectangle_perimeter(10, 15))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS POSTMAN CONSOLE

```
PS C:\Users\katta\OneDrive\Desktop\AIAC> & C:/Users/katta/anaconda3/python.exe c:/Users/katta/OneDrive/Desktop/AIAC/la
Area of Rectangle: 50
Perimeter of Rectangle: 30
Area of Rectangle: 84
Perimeter of Rectangle: 38
Area of Rectangle: 150
Perimeter of Rectangle: 50
```

<u>Task Description #2 (Refactoring – Optimizing Loops and Conditionals)</u>

• Task: Use AI to analyze a Python script with nested loops

and complex conditionals.

• Instructions:

o Ask AI to suggest algorithmic improvements (e.g.,

replace nested loops with set lookups or

comprehensions).

o Implement changes while keeping logic intact.

o Compare execution time before and after refactoring.

• Sample Legacy Code:

```
# Legacy inefficient code
names = ["Alice", "Bob", "Charlie", "David"]
search_names = ["Charlie", "Eve", "Bob"]
for s in search_names:
found = False
for n in names:
if s == n:
```

found = True

if found:

print(f"{s} is in the list")

else:

print(f"{s} is not in the list")

- **Expected Output:**

o Optimized code using set lookups with performance

comparison table.

<u>PROMPT:</u>

#Refactor this Python code to optimize nested loops: replace inner loops with set lookups or comprehensions while keeping the logic identical. Include a simple timing comparison before and after refactoring.

<u>GIVEN CODE AND OUTPUT:</u>

```
#Refactor this Python code to optimize nested loops: replace inner loops with set lookups or comprehensions
"""names = ["Alice", "Bob", "Charlie", "David"]
search_names = ["Charlie", "Eve", "Bob"]
for s in search_names:
    found = False
    for n in names:
        if s == n:
            found = True
            break
    if found:
        print(f"{s} is in the list")
else:
    print(f"{s} is not in the list")"""
import time
names = ["Alice", "Bob", "Charlie", "David"]
search_names = ["Charlie", "Eve", "Bob"]
# Timing before refactoring
start_time = time.time()
for s in search_names:
    found = False
    for n in names:
        if s == n:
            found = True
            break
    if found:
        print(f"{s} is in the list")
    else:
        print(f"{s} is not in the list")
end_time = time.time()
print(f"Time taken before refactoring: {end_time - start_time:.6f} seconds")
# Refactored code using set lookups
start_time = time.time()
names_set = set(names)  # Convert list to set for O(1) lookups
for s in search_names:
    if s in names_set:
        print(f"{s} is in the list")
    else:
        print(f"{s} is not in the list")
end_time = time.time()
print(f"Time taken after refactoring: {end_time - start_time:.6f} seconds")
```

```
Charlie is in the list
Eve is not in the list
Bob is in the list
Time taken before refactoring: 0.001000 seconds
Charlie is in the list
Eve is not in the list
Bob is in the list
Time taken after refactoring: 0.000000 seconds
```

<u>Task Description #3 (Refactoring – Extracting Reusable Functions)</u>

• Task: Use AI to refactor a legacy script where multiple calculations are embedded directly inside the main code block.

• Instructions:

o Identify repeated or related logic and extract it into reusable functions.

o Ensure the refactored code is modular, easy to read, and documented with docstrings.

• Sample Legacy Code:

```
# Legacy script with inline repeated logic

price = 250

tax = price * 0.18

total = price + tax

print("Total Price:", total)

price = 500

tax = price * 0.18

total = price + tax

print("Total Price:", total)
```

• Expected Output:

o Code with a function calculate_total(price) that can be reused for multiple price inputs.

<u>PROMPT:</u>

Refactor this Python code by extracting repeated calculations into a reusable function with docstrings. Make the code modular, readable, and allow multiple inputs without duplicating logic.

<u>GIVEN CODE AND OUTPUT:</u>

```python
#Refactor this Python code by extracting repeated calculatic

def calculate_total_price(price):
    """
    Calculate the total price including tax.
    Args:
        price (float): The base price.
    Returns:
        float: The total price including tax.
    """
    tax = price * 0.18
    total = price + tax
    return total
# Example usage
prices = [250, 500]
for price in prices:
    total_price = calculate_total_price(price)
    print(f"Total Price for {price}: {total_price}")
```

```
ROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   GITLENS   POSTMAN C

S C:\Users\katta\OneDrive\Desktop\AIAC> & C:/Users/katta/anaconda3/pyt
Total Price for 250: 295.0
Total Price for 500: 590.0
```

<u>Task Description #4 (Refactoring – Replacing Hardcoded Values with Constants)</u>

• Task: Use AI to identify and replace all hardcoded "magic numbers" in the code with named constants.

• Instructions:

o Create constants at the top of the file.

o Replace all hardcoded occurrences in calculations with these constants.

o Ensure the code remains functional and is easier to maintain.

• Sample Legacy Code:

```
# Legacy script with hardcoded values

print("Area of Circle:", 3.14159 * (7 ** 2))

print("Circumference of Circle:", 2 * 3.14159 * 7)
```

• **Expected Output:**

o Code with constants like PI = 3.14159 and RADIUS

= 7 used in calculations.

<u>PROMPT:</u>

Refactor this Python code to replace all hardcoded “magic numbers” with named constants at the top of the file. Ensure calculations remain correct and the code is easier to maintain.

<u>GIVEN CODE AND OUTPUT:</u>

```
#Refactor this Python code to replace all hardcoded
PI = 3.14159
RADIUS = 7
print("Area of Circle:", PI * (RADIUS ** 2))
print("Circumference of Circle:", 2 * PI * RADIUS)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS P

```
PS C:\Users\katta\OneDrive\Desktop\AIAC> & C:/Users/katta/anacon
Area of Circle: 153.93791
Circumference of Circle: 43.98226
```

<u>Task Description #5 (Refactoring – Improving Variable Naming and Readability)</u>

• **Task: Use AI to improve readability by renaming unclear**

variables and adding inline comments.

• **Instructions:**

o Replace vague names with meaningful ones.

o Add comments where logic is not obvious.

o Keep functionality exactly the same.

• **Sample Legacy Code:**

```
# Legacy script with poor variable names
a = 10
b = 20
c = a * b / 2
print(c)
```

• **Expected Output:**

o Code with descriptive variable names like base,

height, area_of_triangle, plus explanatory comments.

<u>PROMPT:</u>

#replace vague names with meanningful ones where logic is not obvious and keep functionality exactly the same and add inline comments

<u>GIVEN CODE AND OUTPUT:</u>

<u>Task Description #6 (Refactoring – Removing Redundant Conditional Logic)</u>

• **Task: Use AI to refactor a Python script that contains repeated if–else**

logic for grading students.

• **Instructions:**

o Ask AI to identify redundant conditional checks.

o Replace them with a reusable function.

o Ensure grading logic remains unchanged.

• **Code:**

marks = 85

if marks >= 90:

print("Grade A")

elif marks >= 75:

print("Grade B")

else:

print("Grade C")

marks = 72

if marks >= 90:

print("Grade A")

elif marks >= 75:

print("Grade B")

else:

print("Grade C")

• Expected Output:

o A reusable function like calculate_grade(marks) with clean logic and

docstring.

<u>PROMPT:</u>

#Identify redundant conditional checks and replace them with a reusable function and ensure grading logic remains unchanged

<u>GIVEN CODE AND OUTPUT:</u>

```
lab5.py > ...
#Identify redundant conditional checks and replace them with a reusable function and ensure grading logic remains unchanged
def calculate_grade(marks):
    if marks >= 90:
        return "Grade A"
    elif marks >= 75:
        return "Grade B"
    else:
        return "Grade C"

marks = 85
print(calculate_grade(marks))
marks = 72
print(calculate_grade(marks))
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

SyntaxError: invalid syntax
(base) PS C:\Users\thota\OneDrive\Desktop\AIAC> & C:\Users\thota\anaconda3\python.exe c:/Users/thota/OneDrive/Desktop/AIAC/lab5.py
Grade B
Grade C
```

<u>Task Description #7 (Refactoring – Converting Procedural Code to Functions)</u>

• Task: Use AI to refactor procedural input–processing logic into

functions.

• Instructions:

o Identify input, processing, and output sections.

o Convert each into a separate function.

o Improve code readability without changing behavior.

• Sample Legacy Code:

num = int(input("Enter number: "))

square = num * num

print("Square:", square)

• Expected Output:

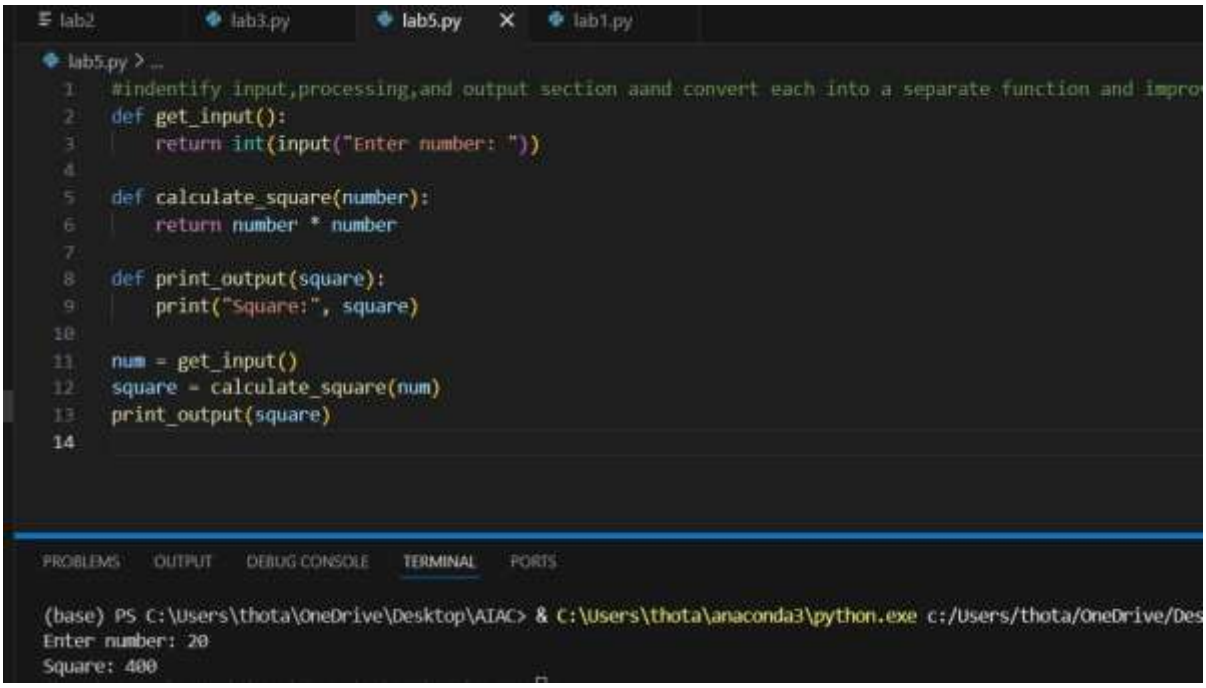
o Modular code using functions like get_input(), calculate_square(),

and display_result().

<u>PROMPT:</u>

#indentify input, processing, and output section and convert each into a separate function and improve code readability without changing behaviour.

<u>GIVEN CODE AND OUTPUT:</u>



<u>Task Description #8 (Refactoring – Optimizing List Processing)</u>

• Task: Use AI to refactor inefficient list processing logic.

• Instructions:

o Ask AI to replace manual loops with list comprehensions or built-in

functions.

o Ensure output remains identical.

• Sample Legacy Code:

nums = [1, 2, 3, 4, 5]

squares = []

for n in nums:

squares.append(n * n)

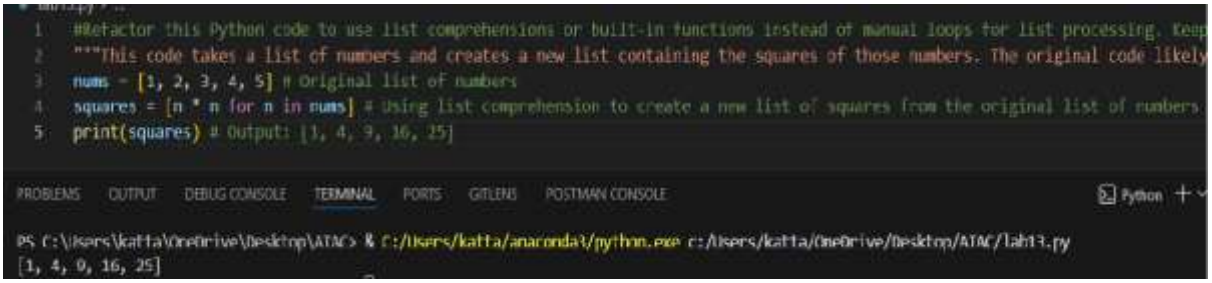
print(squares)

• Expected Output:

o Optimized version using list comprehension with improved

readability

<u>PROMPT:</u>

Refactor this Python code to use list comprehensions or built-in functions instead of manual loops for list processing. Keep the output identical but make the code more concise and readable.

<u>GIVEN CODE AND OUTPUT:</u>